# Who Wrote this Lied?
# A Computational Approach to Lied Classification

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Analysis of symbolic music and computational musicology
Final Project

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# 1. Introduction & Motivation (i)

- ***Lied*:** art song for voice and piano (German: *song)*, based in poetry (Goethe, Heine) themes of nature, dreams, and love

- ***Core Figures:*** (Schubert, Schumann, Brahms) reached its peak in the 19th century

Meine Ruh ist hin, mein Herz ist schwer,
Ich finde sie nimmer und nimmermehr.

Wo ich ihn nicht hab, ist mir das Grab,
Die ganze Welt ist mir vergällt.
Mein armer Kopf ist mir verrückt,
Mein aremer Sinn ist mir zerstückt.

Nach ihm nur schau ich zum Fenster hinaus,
Nach ihm nur geh ich aus dem Haus.
Sein hoher Gang, sein' edle Gestalt,
Seines Mundes Lächeln, seiner Augen gewalt,
Und seiner Rede Zauberfluss,
Sein Händedruck, und ach, sein Kuss!

My peace is gone, my heart is heavy,
I will never, ever find it.

When I don't have him, life's like the grave
The whole world is bitter to me.
My poor head is crazed,
My poor sense is torn to pieces.

I only look out the window for him,
I only leave the house after him.
His high gait, his noble form,
The smile of his mouth, the power of his eyes,
And the magic flow of his words,
His handshake and, oh, his kiss!

**Figure 2** *Excerpt from Goethe's text*

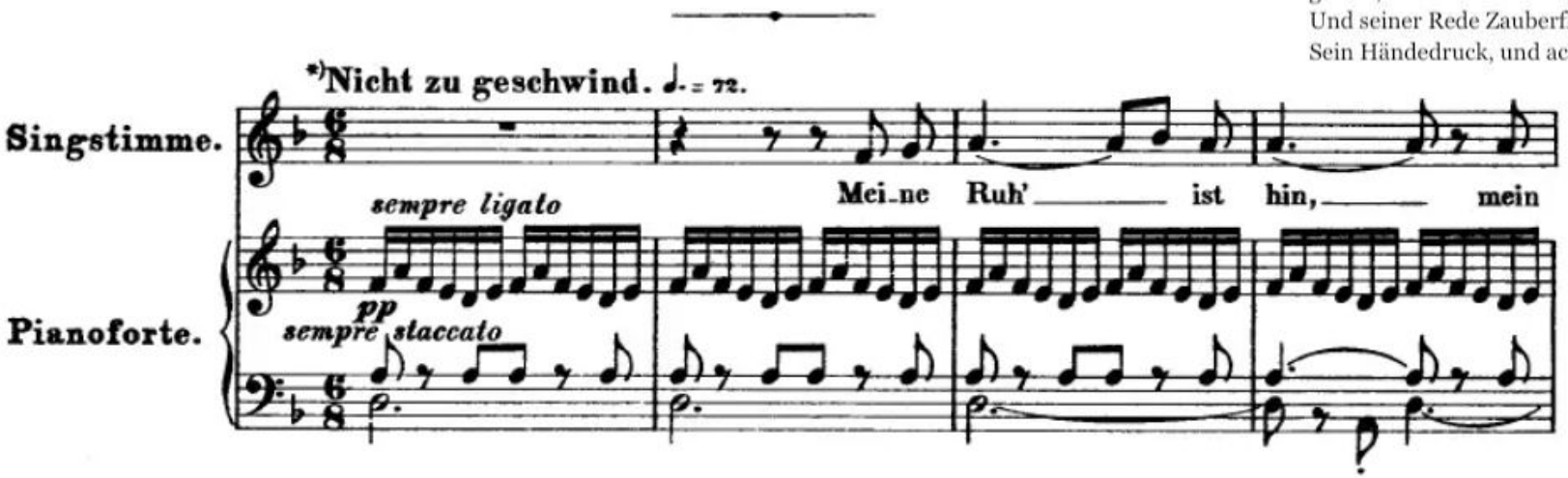


**Figure 1** *Excerpt from Schubert's "Gretchen am Spinnrade", op. 2, D.118*

# 1. Introduction & Motivation (ii)

- **A Classic MIR Task**: Composer Identification.

- **The Challenge**: Cross-era classification is highly successful,

  but same-era and same-genre classification remains remarkably difficult.

- **The Gap**: *Lieder* have received little computational attention for composer attribution - Henzel et al. (2025), Borsan (2021;2025).

# 2. Dataset

- **OpenScore Lieder Corpus** (1,222 *Lieder*) (Gotham & Jonas, 2022)
- **Schubert, Schumann, Brahms** (264 *Lieder)*

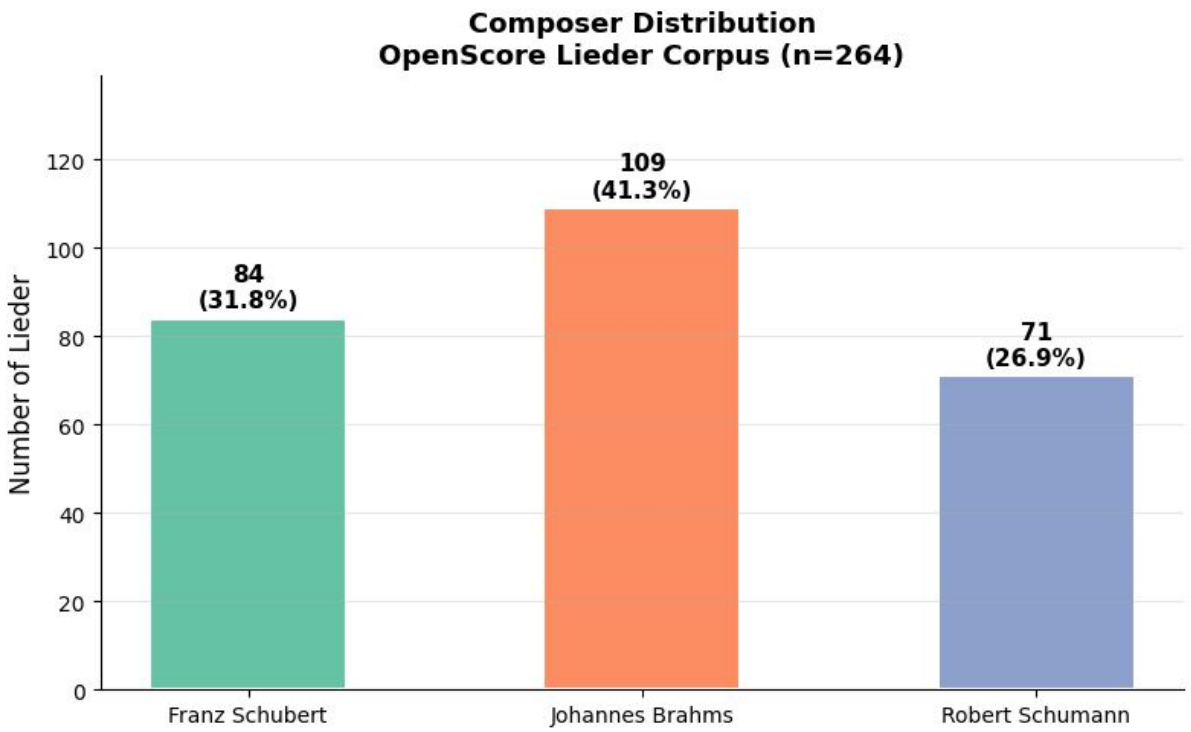


**Figure 2** *Composer distribution in the dataset*

# 3. Exploratory Data Analysis (EDA): Musical Properties (i)

- **Musical Properties**: Time signatures, piece length, and vocal ranges overlap heavily among the three composers.

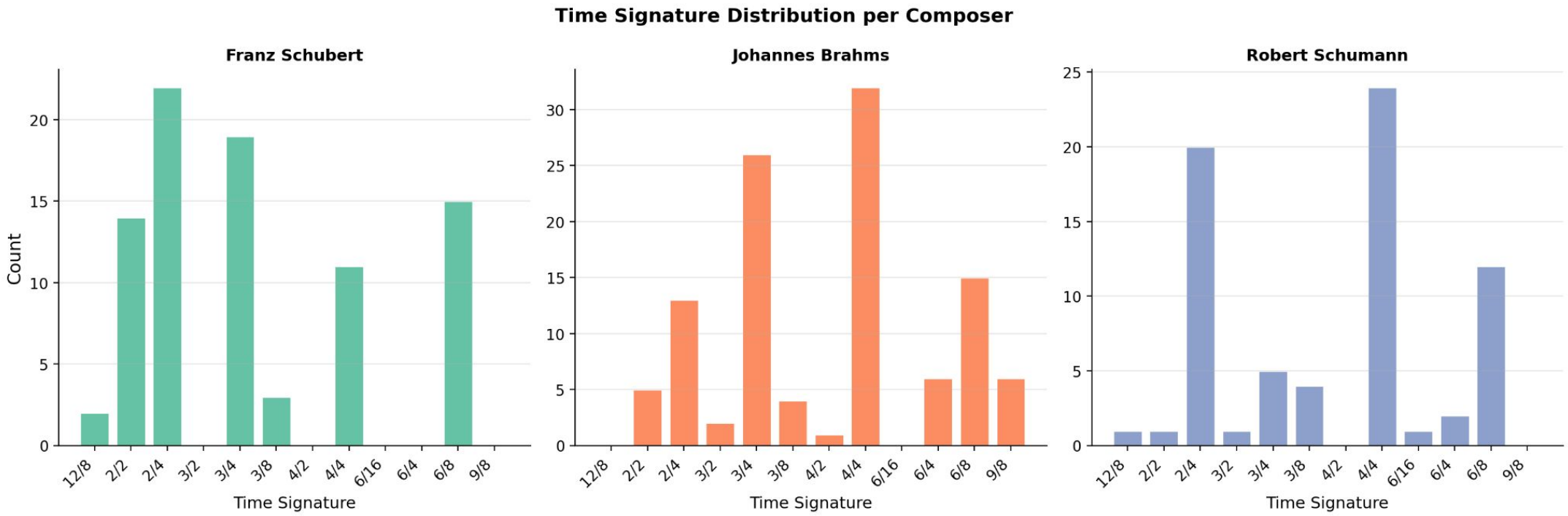


**Figure 3** *Time signature distribution in the dataset*

# 3. Exploratory Data Analysis (EDA): Musical Properties (ii)

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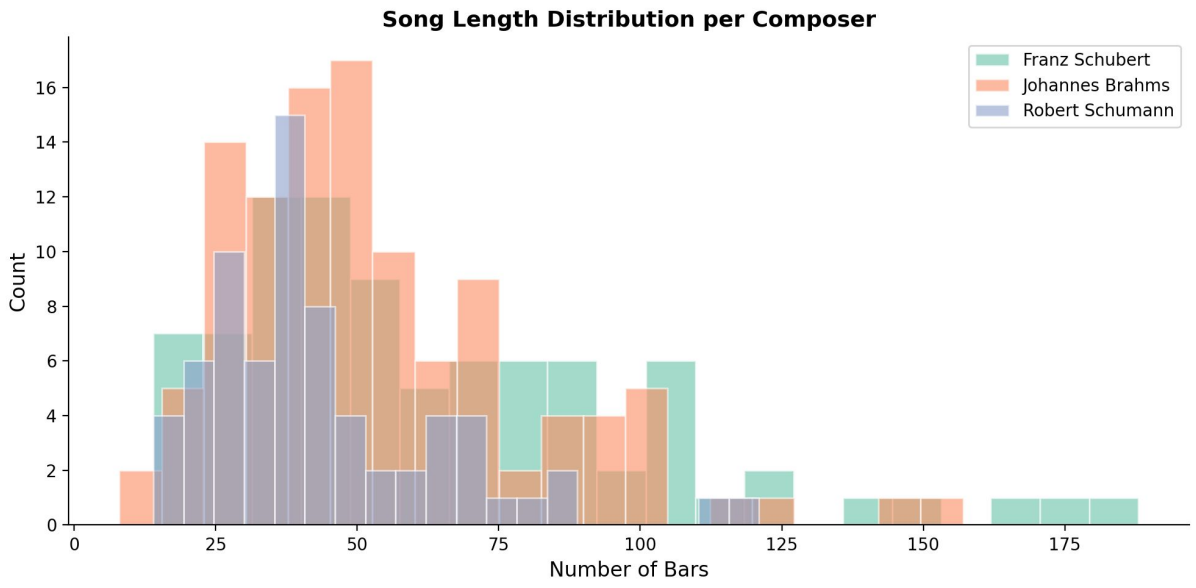


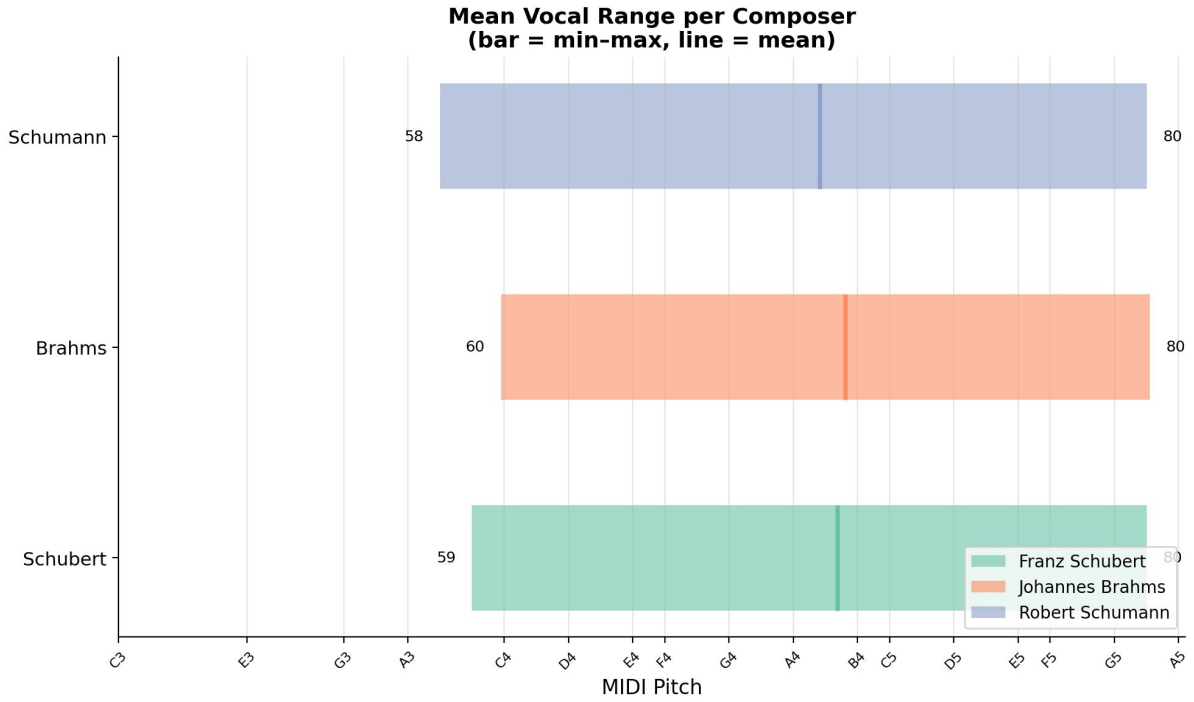


**Figures 4, 5** *Bar length and vocal range distribution for each composer*

# 3. Exploratory Data Analysis (EDA): Text Analysis (i)

**TF – IDF (Term Frequency – Inverse Document Frequency)** – a numerical statistic for most common words across *lieder*

*Total German stopwords: 232*
*=== TOP WORDS PER COMPOSER (TF-IDF) ===*
*Franz Schubert:*
*bächlein, gern, grün, haus, herz, nacht, schlaf, tränen, welt, wohl*
*Johannes Brahms:*
*brust, herz, kind, komm, lieb, liebe, mehr, nacht, traum, wohl*
*Robert Schumann:*
*blumen, herz, himmel, lass, lieb, liebe, mutter, traum, weit, wohl*

**Table 1** *Most repeated words for each composer*

*das Bächlein -small stream*
*grün -green*

*das Herz -heart*
*die Liebe -love*

*die Blumen -flowers*
*der Himmel -sky*

# 3. Exploratory Data Analysis (EDA): Text Analysis (ii)

**LDA (Latent Dirichlet Allocation)** – a statistical model for hidden semantic structures

- Identified 4 topics
- Distributions nearly identical across composers.

*Night & Longing: komm, nacht, wohl, traum, rauschen, hast, ferne, liebe, jäger, lass*
*Sleep & Spirituality: herz, wohl, nacht, schlaf, maria, weit, voll, welt, hand, traum*
*Love & Emotions: liebe, wald, brust, tränen, mehr, welt, himmel, lust, lass, mutter*
*Folk & Nature: gern, grün, liebst, schatz, leben, steh, mehr, möcht, süße, weit*

*die Nacht -night*
*der Traum -dream*

*der Schlaf - sleep(n)*
*das Welt -world*

*tränen -tears*

=== TOPIC DISTRIBUTION PER COMPOSER (n=4) ===

| topic label<br>composer | Night & Longing | Sleep & Spirituality | Love & Emotions | Folk & Nature |
|---|---|---|---|---|
| Franz Schubert | 0.21 | **0.31** | 0.26 | 0.22 |
| Johannes Brahms | 0.30 | **0.36** | 0.25 | 0.08 |
| Robert Schumann | 0.25 | 0.25 | **0.34** | 0.15 |

**Table 2** *Topic distribution for each composer*

# 3. Exploratory Data Analysis (EDA): Text Analysis (iii)

- Distributions nearly identical across composers.

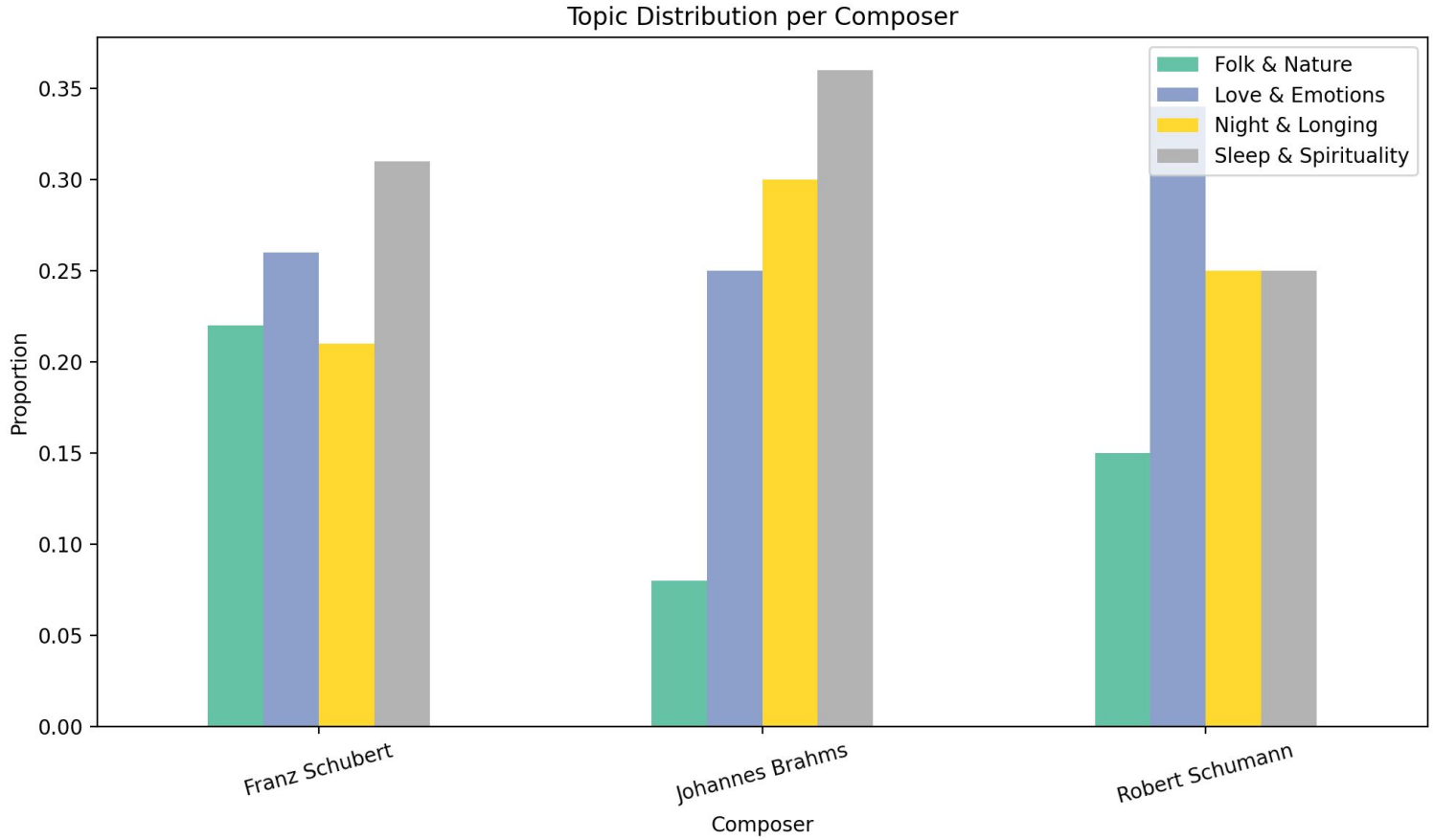


**Figure 6** *Topic distribution for each composer*

# 3. Exploratory Data Analysis (EDA): Text Analysis (iv)

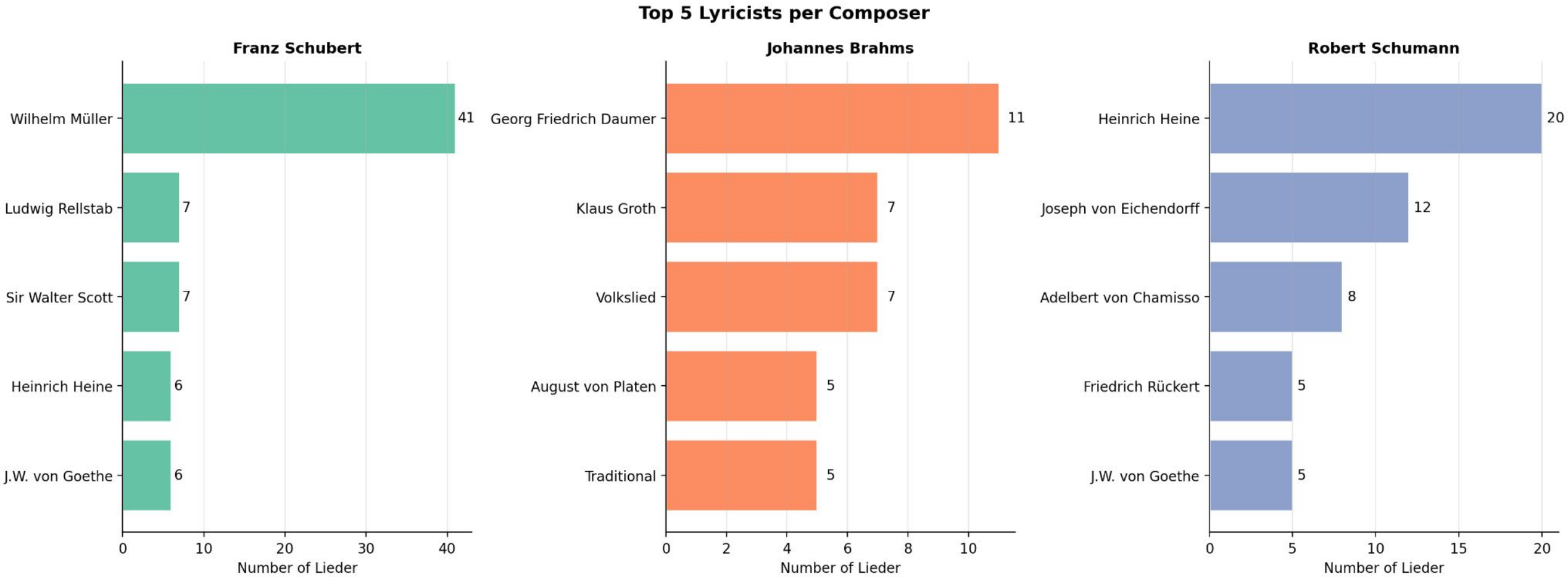


**Figure 7** *Lyricists distribution for each composer*

# 3. Exploratory Data Analysis (EDA): Text-Music Alignment (i)

**Text-Music Alignment**: distinct "text painting" habits

=== TOP WORDS AT HIGH PITCH (>75 MIDI) ===
Franz Schubert:
**herz**, **ruh**, **tränen**, gern, liebst, süsse, **maria**, bächlein, reissend, träume
Johannes Brahms:
**liebe**, **herz**, **seele**, gar, liebsten, schönen, mehr, **ach,** himmel, lieb
Robert Schumann:
**weh**, **liebe**, jemand, **ferne**, ganz, lied, wildnis, wüsstens, drein, flüstern

=== TOP WORDS AT LOW PITCH (<60 MIDI) ===
Franz Schubert:
rauhen, reicht, audige, helle, dach, duft, ben, eibus, bet, ner
Johannes Brahms:
**tod**, wohl, wäre, **stirbt**, **odem**, **mensch**, sonne, tust, hätte, wann
Robert Schumann:
sei, tag, jagt, alte, wohl, schar, wach, munter, wohin, wild

**Table 3** *Word-pitch pairs for each composer*

*die Ruh -quietness*
*tränen -tears*
*Maria -The Virgin Mary*

*die Liebe -love*
*das Herz -heart*
*die Seele -soul*

*das Weh -pain*
*die Ferne -distance*

*der Tod -death*
*stirbt -die(third person)*
*der Mensch -man*

# 4. Methodology : Domain-Driven Handcrafted Features (i)

To capture subtle compositional signatures without overwhelming the classifier, we designed 4 low-dimensional features, summarized by 3 statistical descriptors (mean, std, entropy) into a **12D vector**:

1. **Tonal Tension**: Euclidean distance of chords from the tonal center (Spiral Array Model).
2. **Harmonic Complexity**: Shannon entropy of the pitch class distribution per beat.
3. **Pianistic Texture**: Onset density in the accompaniment (distinguishes thick block chords from sparse arpeggios).
4. **Melodic Contour**: Categorical encoding of vocal interval motion (leap, step, unison, downward and upward).

# 4. Methodology : Domain-Driven Handcrafted Features (ii)

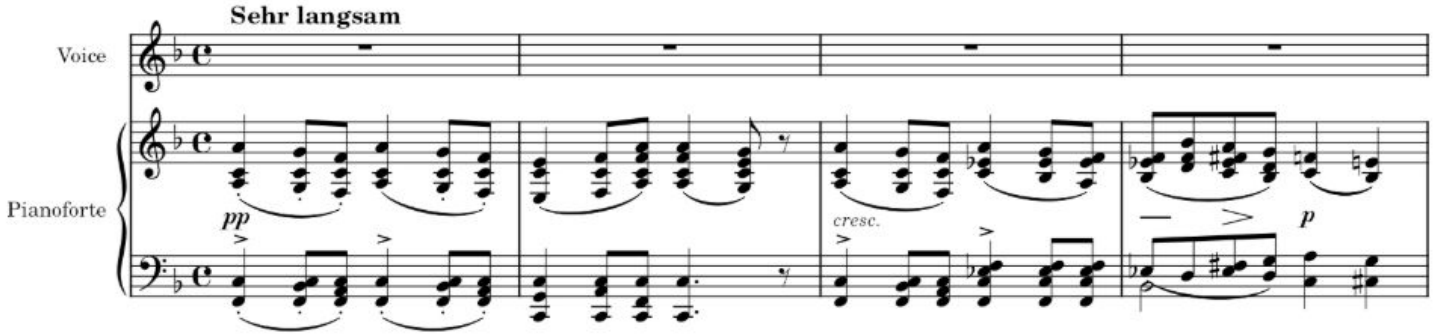


**Figure 8** *Excerpt from Franz Schubert's "Winterreise" Cycle, D.911. no 21 "Das Wirtshaus"*

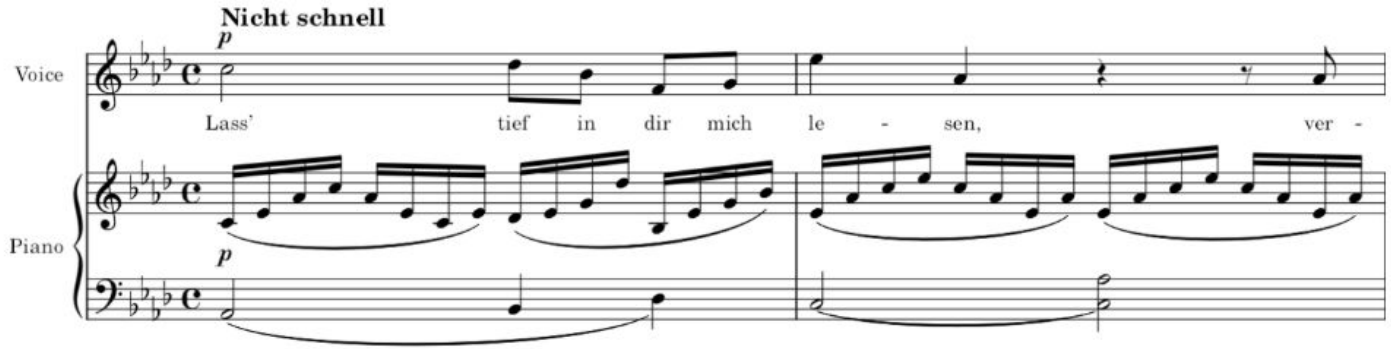


**Figure 9** *Excerpt from Robert Schumann's "Lieder und Gesänge", Vol.IV, op.96, no.3 "Ihre Stimme"*

# 5. Deep Learning Baseline: Adversarial-MidiBERT

- **State-of-the-Art Performance**: A BERT-based symbolic music model pre-trained on massive MIDI datasets (>15,000 compositions).
- **Our Implementation**: We utilized this highly capable pre-trained model as a feature extractor, generating **768-dimensional deep embeddings** for each Lied to feed into our downstream classifiers.

| Model | Pre-train | | | Sequence-Level Classification | | Token-Level Classification | |
|---|---|---|---|---|---|---|---|
| | Accuracy | Epochs | Time | Composer | Emotion | Velocity | Melody |
| MidiBERT [4] | 79.60% | 500 | 6.44d | 79.07% | 67.59% | 44.88% | 92.53% |
| MusicBERT-QM [20] | 80.57% | 500 | 9.47d | 83.72% | 69.52% | 46.71% | 92.64% |
| MusicBERT [3] | 76.01% | 500 | 10.06d | 86.05% | 71.06% | 38.79% | 92.47% |
| PianoBART [5] | **96.67%** | **268** | **3.19d** | 88.37% | 73.15% | **49.37%** | 92.62% |
| Adversarial-MidiBERT (ours) | 81.47% | 436 | 9.82d | **97.92%** | **79.46%** | 45.58% | **92.68%** |
| Adversarial-MidiBERT (fine-tune w/o mask) | 81.47% | 436 | 9.82d | 65.98% | 70.53% | 45.30% | 92.55% |
| Adversarial-MidiBERT (w/o pre-train) | – | – | – | 79.76% | 68.75% | 38.70% | 87.98% |

**Table 4** *Model Performance in Different Tasks: The bold and underlined value indicates the best and second best result within each task*

## 6. Methodology : Model Details

- **Deep Embeddings**: Extracted **768D** semantic features using Adversarial-MidiBERT.
- **Classifier Configurations**:
  i. **SVM**: Trained on the 12D handcrafted features.
  ii. **SVM**: Trained on the 768D MidiBERT embeddings.
  iii. **2-layer MLP** (128->64 neurons, L2 regularization): Trained on 768D MidiBERT.
- **Evaluation**: Rigorous Stratified 5-fold Cross-Validation.

## 7. Sanity Check: Is the model broken, or is the task hard?

If accuracy is low, is the pipeline flawed? To test this, we ran the same baseline (MidiBERT + SVM) on a stylistically distinct early music dataset:

- **Test Data**: PMB dataset (Palestrina: 100, Monteverdi: 97, Bach: 100).
- **Results**: Balanced Accuracy of **98.96%** (F1-scores > 0.98 for all classes).

**The classification pipeline works perfectly**

## 8. Results

On the 264-song Lied dataset, **low-dimensional handcrafted features outperformed high-dimensional deep embeddings**:

| Feature Config | Classifier | Dimensions | Balanced Accuracy |
|---|---|---|---|
| **Ours** | **SVM** | **12D** | **0.4935** (std: 0.0465) |
| MidiBERT | SVM | 768D | 0.4709 (std: 0.0252) |
| MidiBERT | MLP | 768D | 0.4528 |

*(Note: All results are significantly above the 0.33 random baseline)*

**Table 5** *Classification results with different models*

# 9. Discussion

1. **The Robustness of Domain Knowledge**: In small-sample scenarios with highly overlapping styles, 768-dimensional pretrained embeddings suffer from task misalignment due to excessive irrelevant noise, weakening their ability to distinguish fine-grained composer styles; in contrast, low-dimensional handcrafted features focused on core musicological intuition demonstrate greater robustness and effectiveness.
2. **The Stylistic Bottleneck**: All configurations achieve accuracies between 0.45 and 0.50 on the Lied dataset. Combined with MidiBERT's accuracy of 98.96% on the PMB dataset, this confirms that the current performance bottleneck stems not from feature extractor or classifier, but from the extreme stylistic similarity among composers sharing the same genre and historical period.

## 10. Core Contributions

1. **Defining a highly challenging MIR benchmark**

   Same-era Romantic *Lied* classification is incredibly difficult.

   A pipeline that scores ~99% on Baroque data drops to ~49% here, highlights the extreme intra-genre convergence.

2. **Proving the value of handcrafted features in data-constrained tasks**

   In tasks demanding high domain expertise but lacking massive datasets, lightweight and interpretable features (12D) are not just computationally cheaper.

   They can rival or exceed large pre-trained deep learning embeddings (768D).

# Thank you for listening!

Q & A